AI SMART ROUTE FINDER

Artificial Intelligence Project Report

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Course	Artificial Intelligence
Project	AI Smart Route Finder

1. Introduction

Artificial Intelligence provides techniques for solving problems by representing states, actions and goals and then searching for suitable solutions. This project implements an AI-based Smart Route Finder in a grid environment. The application allows a user to create obstacles and observe how different search algorithms explore the grid and find a route from a start position to a destination.

2. Problem Statement

The objective is to design a route-finding system that can search a grid containing obstacles and determine a path between a start point and a goal point. The project compares multiple AI search strategies and visualizes their behavior.

3. Objectives

• Implement classical AI search algorithms in a practical application.
• Represent route finding as a state-space search problem.
• Allow obstacles to be placed in the environment.
• Visualize the exploration process and final route.
• Compare algorithms using path length, nodes explored and execution time.

4. Algorithms Implemented

The project implements Breadth First Search (BFS), Depth First Search (DFS), Greedy Best First Search, A* Search and Hill Climbing. Each algorithm uses a different strategy for deciding which state should be explored next.

5. A* Search

A* Search is the primary informed search technique demonstrated in the project. It evaluates a node using **$f(n) = g(n) + h(n)$**. $g(n)$ is the cost from the start state to the current state and $h(n)$ is the estimated cost from the current state to the goal. Manhattan Distance is used as the heuristic.

6. State-Space Representation

Each grid cell is treated as a state. The start cell is the initial state, the destination is the goal state, and moving to an adjacent free cell represents an action. Obstacle cells are not available for traversal.

7. Technologies Used

• Python
• Tkinter GUI
• Queue, stack and priority-queue based data structures
• Classical Artificial Intelligence search techniques

8. System Features

The application provides a graphical grid, obstacle creation, destination selection, algorithm selection, route visualization and performance comparison. It is designed to make AI search behavior easy to observe.

9. Algorithm Comparison

BFS generally finds a shortest path when every movement has equal cost, while DFS focuses on depth and may return a longer path. Greedy Best First Search prioritizes states that appear close to the goal. A* combines actual path cost with a heuristic estimate. Hill Climbing attempts to move toward a better-looking state and can become trapped by local conditions.

10. Applications

The ideas demonstrated by this project are relevant to robot navigation, game pathfinding, map routing, autonomous systems and other problems where an agent must search through possible states to reach a goal.

11. Future Scope

The project can be extended with weighted terrain, diagonal movement, dynamic obstacles, larger maps, additional heuristics, Dijkstra's algorithm, bidirectional search and animated performance comparison charts. A web-based version could also be developed.

12. Conclusion

The AI Smart Route Finder demonstrates how classical search algorithms can be applied to a real route-finding problem. By visualizing the search process and comparing strategies, the project provides practical understanding of state-space representation, uninformed search and informed search, especially A* Search.